

Algorithmic Complexity vs. Market Efficiency: Evaluating Wavelet–Transformer Architectures for Cryptocurrency Price Forecasting

Aldan Jay, Rafael Berlanga — Universitat Jaume I. Algorithms 2026, 19(2), 101. DOI 10.3390/a19020101. Open Access. (Text extract of the published open-access article.)

Abstract

We investigate whether sophisticated deep learning architectures justify their computational cost for short-term cryptocurrency price forecasting. Our study evaluates a 2.1M-parameter wavelet-enhanced transformer that decomposes the Fear and Greed Index (FGI) into multiple timescales before integrating these signals with technical indicators. Using Diebold–Mariano tests with HAC-corrected variance, we find that all models — including our wavelet–transformer, ARIMA, XGBoost, LSTM, and vanilla Transformer — fail to significantly outperform the $O(1)$ naive persistence baseline at the 1-day horizon (DM statistic = +19.13, $p < 0.001$, naive preferred). Our model achieves an RMSE of USD 2005 versus USD 1986 for naive (ratio 1.010), requiring $3909\times$ more inference time (2.43 ms vs. 0.0006 ms) for statistically worse performance. These results provide strong empirical support for the Efficient Market Hypothesis in cryptocurrency markets. Through systematic ablation, we identify positional encoding as the only critical architectural component — its removal causes 30% RMSE degradation.

Data and method

BTC, ETH, BNB daily OHLCV from Binance and Coinbase (1 Feb 2018 – 30 Sep 2025, 2794 records each) plus the Alternative.me Crypto Fear and Greed Index. The FGI is decomposed with a 4-level Daubechies-4 DWT into five bands (D1–D4, A4), each with a 60-day lookahead, combined with 12 technical indicators. Walk-forward temporal 60/20/20 split. Crucially, a leakage-safe rolling decomposition is used: for each timestamp t the db4 transform is computed on the last 60 observed values only, with symmetric padding within the window, so no future observation influences any coefficient used to predict at t .

Key results

No model significantly beats naive persistence; RMSE ratios fall within 1.003–1.065 and all achieve ~50% directional accuracy. Wavelet–transformer RMSE USD 2005 (ratio 1.010) beats vanilla Transformer (USD 2098) and XGBoost (USD 2116), matching LSTM. Ablation: removing positional encoding degrades RMSE by 30.2% (USD 2005 → USD 2611); cross-attention, skip connections and encoder depth are negligible ($p > 0.05$). Multi-seed CV 0.77%. The DM statistic favouring naive falls from 19.13 (1-day) to 6.81 (30-day). Cross-asset transfer fails catastrophically (ETH RMSE USD 421,600; BNB USD 2.8M) because the FGI is Bitcoin-centric. Conclusion: strong support for weak-form efficiency at ≤ 30 -day horizons; architectural complexity gives negative ROI. All code and data are public.